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1 Introduction

Air travel has increased considerably in recent years. Today, airline problems are based on the competitive morality and the pricing pressure. The companies have to try their best to reduce their costs. So apart from consideration of the passengers' wait time, from the airline's point of view, time is money. Minimizing the turnaround time is very important. "An airplane only generates revenue when it is in the air," explains David Swierenga, a former Air Transport Association of America economist and president of aviation consulting company Aeroecon. And long boarding times limit the number of trips that a plane can make in a day.

Many factors contribute to turnaround time, including passenger boarding, baggage handling, refueling and aircraft cleaning. The passenger boarding is the part that takes the longest time and therefore plays the most significant role. This is studied by Boeing 1998. Then, different solutions have been proposed to reduce the boarding time and there a lot of airlines using different methods. The main goal is to get the passengers sitting on their favorite seat in a fast and efficient way.

Van Landeghem and Beuselinck (2002)^[1] identify that the typical aircraft turn-around (coming to and leaving the gate) is approximately 30-60 minutes. Airlines seem to adopt different aircraft boarding strategies based on airline culture and service level (Van de Briel et al., 2005)^[2]. Some airlines do not impose any strategy and left the passengers board randomly. Others arrange the passengers into groups, zones or call-offs based on specific boarding strategy adopted by the airline. Each of these groups is then called to board the aircraft in sequence.

The most common traditional boarding aircraft strategies are back-to-front and out-in. In both strategies, passengers are typically assigned to groups. First-class and special need passengers are boarded first. In the back-to-front strategy, passengers join the boarding queue, one row at a time. In the back-to-front strategy, as the name implies, passengers start filling up the aircraft from back to the front. The passengers are divided into groups based on the row of their seats from the back of the aircraft to the front. The number of boarding groups should be as small as possible, otherwise, the call-off gets complicated.

These methods provide a good understanding of existing boarding strategies and enable us to evaluate various known strategies. Some reported research and published works on this topic focus on modeling the problem using computer-based simulation (H. Van Landeghem, A. Beuselinck 2001)^[3]. Simulation approaches can help achieve strategies, but the conclusion in the form of mathematics is not gained.

A recent and very interesting approach to aircraft boarding strategy is by Massoud Bazargan^[4]. He model the aircraft boarding strategy using a mixed integer linear program. The model attempts to minimize the total interferences among the

passengers .

This paper presents a new multi-goal program to generate efficient boarding strategies . The mathematical model attempts to minimize the boarding time and the total interferences among the passengers which are major causes for boarding delays. The model provides the flexibility to optimize the boarding times and the interferences . The model are also allows us to examine the interval time a passenger is allowed to board . This paper then adopts the mathematical model to a single aisle Airbus-320 aircraft . The midsize boeing 767-400 and the large boeing 747-400 are also analysis in this paper .

2 General Assumptions

- Airplanes are 100% load. There are no empty seats.
- We ignore the accidents happening in the airplane. Sometimes accidents will happen in the airplane. For example, a passenger may tumble in the aisle, which will greatly block the aisle. As the probability of an accident is low, we ignore it.
- Passengers with special needs board first for any a boarding strategy. Therefore their influence to every strategy is the same. So we ignore them in our simulation.
- In the airplane seats are grouped. Seats in the same group have the same color. This is similar to assigning passengers to groups.
- A passenger in group k sitting on a seat is a stochastic process. For example, if a passenger in group k is assigned to block 3, then he can choose any a seat in block3 when he reach block 3.

For small airplane

- It has one boarding gate. That is, all the passengers must pass through the gate if they want to get into the airplane;
- It has only one aisle.
- One row has 6 seats and the two sides of the aisle have the same seats.

For middle airplane

- It has one boarding gate.
- It has two aisles and every aisle have the same people at any time. If a passenger gets through the one, the one after the passenger gets through the other one.

For large airplane

- It has two floors and two boarding gates, one for the main deck, the other for the upper deck.
- Passengers between different decks don't interference each other.
- Boarding time for the main deck is larger than the upper deck. Many large airplanes have much more seats in the main deck. For example, an airbus with 550 seats has 360 seats in the main deck and 190 seats in the upper deck.

3 Variable notation

Table 1 Variables used in the model

Notation	Description	Units
n	row number in the airplane	
Q	total interferences time	s
T	total boarding time	s
t_1	the time when the first passenger boarding the plane	s
t_2	the time when the last passenger is seated	s
α	“pass coefficient” for seat interferences	
β	“pass coefficient” for aisle interferences	
(i,j)	a seat in row i, column j	
y_1	per seat interference time, y_1 is a random variable	s
y_2	per aisle interference time, y_2 is a random variable	s
G	number of groups	
TS	total seat interferences time	s
AW	total aisle interferences time within groups	s
AB	total aisle interferences time between groups	s
ω_1	weight of interferences time	
ω_2	weight of boarding time	

4 Analysis of boarding system

Why should anyone be concerned about how passengers are boarded onto an airplane? As mentioned before, the more time that a plane has to spend on the tarmac, the less profit is earned by the airline operating that aircraft. And the another answer to this question may come easily to those who have traveled frequently on international or cross-continental flights. Boarding orders are important in that they determine passenger satisfaction and safety, as these factors affect the image of the

airline .

Combining the information in internet and that the problem has just been presented , we can get

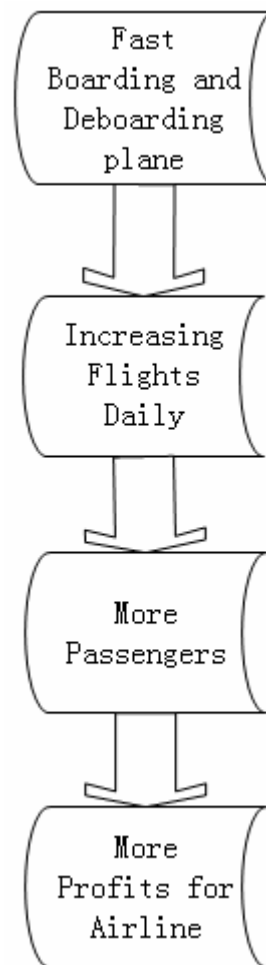


Figure 1 How turnaround time effect the airline profitability

4.1 Boarding system

We compare the four boarding strategies have been presented

- Reverse pyramid system: rear window and middle first, front window and middle next, followed by rear aisle, then front aisle.
- Rotating zone system : Customers get assigned to Group A , B or C on their boarding passes , in the order in which the passenger checks in . Groups are called in alphabetical order , with passengers rushing to occupy the seat of their choice .
- Outside-in system : for window first, then middle and then aisle, a technique favored by Delta and United
- Back-to-front : Passengers are seated by groups of rows, beginning with the row at the back of the plane and proceeding forward.

A simple comparison between them :

Back-to-front and Rotating zone system: We compare them by using the

figure in the website given by the problem. In step 3 and 5, Rotating zone system is more efficient than Back-to-front, while in step 4 and 6 it is less efficient. In step 3, Rotating zone system will encounter less aisle interferences as the two groups hardly interfere each other. However, in step 4, the group in step 3 may hold back the next group for reaching the middle block, while this influence may be much smaller in Back-to-front. From the two figures we know 3Back-to-front is better than Rotating zone system, we think this is because the Rotating zone system's aisle interferences are greater than Back-to-front.

Outside-in and reverse pyramid: Both of the two boarding strategies work very well. Compared to the two boarding strategies of Rotating zone system and Back-to-front, they have less seat and aisle interferences. We can even say that seat interferences hardly happen. But we see from the figure that Outside-in is better than reverse pyramid, we think that the main difference between them are the process of putting baggage. Outside-in is more convenient in putting baggage than reverse pyramid.

4.2 Boarding process

- Passengers are called by the gate agent , announcing the start of boarding . Passengers will typically start queuing at the gate . Delaying factors are missing passengers , that need to be paged , or late arrival of transfer partners .
 - The gate agent controls the boarding pass of each passenger , and registers his entry using a ticket reader . It has become customary to seat passengers with special needs first , followed by first-class passengers . Then coach and business-class passengers are seated by groups of rows , beginning with the row at the back of the plane and proceeding forward .
 - Finally passengers access the airplane through the access bridge and the front door . Once inside the plane , they have to queue in a single line (due to the small aisle width) until they reach their assigned seat .
- In practice, step 1 and 2 don't contribute to the Turn Time . Step 3 plays the very important role in the Turnaround Time .

4.3 Interference

- Aisle interference
When passenger loads luggage into the overhead storage compartments of the aircraft it occurs . And this passenger must stand in the narrow aisle as a obstacle to those waiting to get to their seats . Unless someone can squeeze past .
- Seat interference
When a passenger's route to his or her seat is blocked by another passenger sitting in the same row

It is obvious that if the frequency of interference increases during the boarding

of an aircraft , the boarding time increases . So the method should reduce the interference to minimize the boarding time .

4.4 Definition of parameters

4.4.1 Definition

We should define some parameters about delaying time when a interference occurs .

- Row speed , the speed of passenger traveling from one row to the next , was 1.5 seconds .
- 60% of the passengers have luggage , the speed that passengers place their carry-on luggage into the overhead bins obey a function , with a low of 3 seconds , an average of 7 seconds and a high of 11 seconds .
- As related to luggage speed , a “pass coefficient ” is determined . When a person is loading his or her luggage into the overhead bins , on average , one passenger out of every ten would slide past .
- Seat interference time is found , with an average of 9.7 seconds .

4.4.2 Reasonable analysis

Four boarding systems are presented , and they are simulated with defined parameters above . All first class passengers are to be boarded first in every boarding method . Therefore , this simulate will mainly apply to the boarding of passengers in the economy class .

All simulations are run 1000 times . (See the program in appendix [1]) . The conclusions are summarized as follow :

Table 2 compare the simulation and the experiment conclusion

boarding system			relative error
Outside-in	simulate time	22.8	0.44%
	experiment time	22.9	
Back-to-front	simulate time	24.1	3.20%
	experiment time	24.9	
Rotating-zone	simulate time	24.5	3.54%
	experiment time	25.4	

The simulate time is smaller than the experiment time . Because in actual process , some unpredictable factor may occur . The relative error is small , so the parameters can be used in later model .

5 Outside-in strategy

Many reported research and published works on boarding strategy focus on

modeling the problem using computer-based simulation . From the reference [1] , [2] , [3] , [4] , we can obtain the good boarding procedure : Outside-in , Reverse-pyramid , U1, U2 . In this section , passengers arrive at a constant rate . In reality , this rate is determined by the gating operations .We choose not to include this step in our study , in order not to obscure the pure effect of sequencing . Then, we simulate it (See the program in appendix [2]), and the conclusions are summarized in table 3

Table 3 The minimum time of the four boarding procedure

Methods	Outside-in	Reverse-pyramid	U1	U2
Time(Min)	11.4386	15.7191	11.5052	11.5999

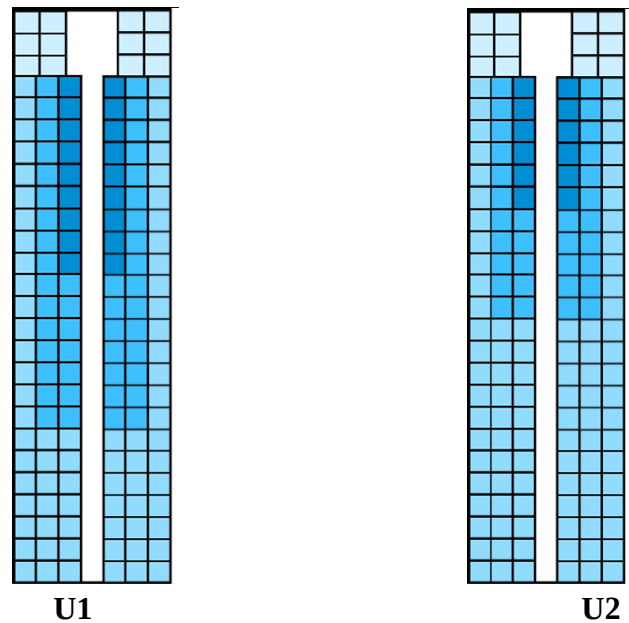


Figure 2 Boarding processes about U1 and U2

Following the conclusion , we can obtain that the Outside-in boarding procedure is the optimal method .

So the average **time interval**, which represents the elapsed time between two consecutive boarding passengers. is denoted by

$$T_{interval} = \frac{11.4386 \times 60}{150} = 4.57s$$

It means that a passenger is allowed to board every 4.57 seconds . Is it the best project? We need to further consideration .

6 Multiple objective programming model

Note that the total boarding time T increases as the passenger interval time increases. And the total interferences time Q will decrease as the passenger time interval increases.(see Fig 7).

Our objective is to find the optimal passenger interval time to minimize the total boarding time and total interferences time.

$$\text{minimize } T \quad (1)$$

$$\text{minimize } Q \quad (2)$$

(1) Total boarding time T

From a micro perspective, a boarding process is very complicated. For example, we should consider the interferences described above to each passenger, and one may choose a wrong seat so he should go back to his seat that may make the aisle situation worse. In other words, the situations inside the airplane during the boarding process are stochastic. Therefore we consider the boarding process from a macro perspective.

The variables that we need to know is the time when the first passenger get into the airplane and the time when the last passenger is seated in his seat. We use t_1 and t_2 to represent them separately. Hence the total boarding time T can be written as follows

$$T = t_2 - t_1$$

(2) Total interferences time Q

$$Q = TS + AW + AB$$

Here TS represent the total seat interferences time, AW represents the total aisle interferences time within groups, AB represents total aisle interferences time between groups. Next we will explain TS, AW, AB separately.

Basing on the linear programming model developed by Massoud Bazargan[4], we develop a new expression for computing interferences time. The work we have done is to simplify and improve his model, and particularly we add a new parameter called “pass coefficient”. That is to say, when a passenger is loading his or her luggage into the overhead bins, the stalled passengers would slide past or “pass” the aisle interference without using much time. We use α to represent it. $\alpha = 0.1$ represents one passenger out of every ten would pass the aisle interference without much time. Similarly, a passenger may also pass seat interference without much time so the seated passenger won’t need to get out to the aisle. We use β to represent it.

$\beta = 0.1$ represents one passenger out of every ten would pass the seat interference without much time.

Each seat in this aircraft is represented by (i, j) where i (i = 1, . . . , n) is the row and j (j=A,B,. . . ,F) is the location of the seat within row i. And passenger (i,j) donate a passenger sit at (i,j).

First we define the following binary decision variable

$$x_{i,j,k} = \begin{cases} 1 & \text{if seat } j \text{ in row } i \text{ is assigned to group } k \\ 0 & \text{otherwise} \end{cases}$$

● Seat interferences time TS

We define

$$S_{i,j,k} = \begin{cases} 1 & \text{if passenger } (i, s) \text{ in row } i \text{ interfere passenger } (i, j) \\ 0 & \text{otherwise} \end{cases}$$

$$s = A, B, C, D, E, F \quad \text{and} \quad s \neq j$$

$S_{i,j,k}$ can be expressed as the following constraint

$$S_{i,j,k} \geq x_{i,j,k} + \sum_{l=1}^k x_{i,s,l} - 1 \quad \forall i, \text{ and } k > 1$$

On the right hand side of this constraint, we have $x_{i,j,k}$ which represents if the passenger (i,j) in the seat (j) boards in group k and $\sum_{l=1}^k x_{i,s,l}$ indicates that if the passenger sitting in the seat (s) has boarded in all previous groups (including k). The term (-1) in the above constraint is added to set the value of $S_{i,j,k}$ equal to 1 if there is an interference and 0 otherwise. Table 4 clarifies this further and examine the values of $S_{i,j,k}$ for a given row i.

Table 4

$x_{i,j,k}$	$\sum_{l=1}^k x_{i,s,l}$	$S_{i,j,k} \geq$	Comments
1	1	1	Interference occurs
1	0	0	No interference
0	1	0	No interference
0	0	-1	No interference

Considering the “pass coefficient”, the seat interferences time for passenger (i,j) can be written as $\alpha \sum_s y_1 S_{i,j,k}$. Let TS represent the total seat interferences time, then

$$TS = \alpha \sum_{i=1}^n \sum_{j=1}^m \sum_{k=2}^G \left(\sum_s y_1 S_{i,j,k} \right)$$

● Aisle interferences

Aisle interferences are composed of within groups aisle interferences and between groups aisle interferences.

● Within groups aisle interferences time AW

Let the integer variable $AW_{i,j,k}$ represent the (maximum) number of aisle interference for passenger (i, j) assigned to group k. Similar to the previous description, we can write the constraint for $AW_{i,j,k}$ as follows:

$$AW_{i,j,k} \geq (6(i-1) + 4)x_{i,j,k} + \sum_{u=1}^i \sum_{v=1}^6 x_{u,v,k} - (6(i-1) + 5)$$

On the right hand side of this constraint, the first term takes a value of $6(i-1)+4$ if the passenger in seat (i, j) is assigned to group k and zero otherwise. Because the next term includes $x_{i,j,k}$, we add 4 rather than 5. The second term, adds up all the passengers in the same group k with passenger (i,j)(including (i,j)). The third term on the right hand side is adopted to provide the correct count on aisle interferences for $AW_{i,j,k}$. Table 5 clarifies this further.

Table 5

$x_{i,j,k}$	$AW_{i,j,k} \geq$
1	$\sum_{u=1}^i \sum_{v=1}^6 x_{u,v,k} - 1$
0	≤ 0

Apparently, the lowest number of aisle interferences for the passenger (i,j) is 0. So we

take number of aisle interferences as $\frac{0 + AW_{imj,k}}{2} = \frac{1}{2} AW_{imj,k}$, the aisle interferences

time for passenger (i,j) is $\frac{1}{2} y_2 AW_{imj,k}$

Then the total aisle interferences time within groups

$$AW = \frac{1}{2} \beta \sum_{i=2}^n \sum_{j=1}^m \sum_{k=1}^G y_2 AW_{i,j,k}$$

● Between groups aisle interferences AB

We define the integer variable $AB_{i,j,k}$ to represent the maximum number of aisle interferences for passenger (i, j) who boards in group k ($k > 1$) with all the passengers in the previous group (k -1). We write the following constraint for $AB_{i,j,k}$:

$$AB_{i,j,k} \geq 6(i)x_{i,j,k} + \sum_{u=1}^i \sum_{v=1}^6 x_{u,v,k-1} - 6(i)$$

The first term on the right hand side of this constraint takes a value of 6i if passenger (i, j) is assigned to group k and 0 otherwise. The second term adds up all the passengers who boarded in group (k -1). Again the third term is used to provide the correct count.

This constraint for $AB_{i,j,k}$ assumes that none of the passengers in the previous group is seated when the passenger in seat (i, j) assigned to group k boards the aircraft. Of course, the expected number of aisle interferences for this passenger depends on how quickly each group of passengers is called to board the aircraft. We will assume that for any passenger in group k ($k > 1$) boarding the aircraft, there is a fraction of passengers from the previous group (k-1) still in the jet-way trying to reach to their rows and seats. We call this fraction λ ($0 \leq \lambda \leq 1$). Therefore, the expected number of aisle interferences between the passenger (i, j) assigned to group k with passengers in group (k -1) is $\lambda AB_{i,j,k}$. Then the interferences time for passenger (i,j)

is $\lambda y_2 AB_{i,j,k}$. When λ is 0, no aisle interferences occur between groups. This occurs when a new group of passengers is called to board the aircraft and all passengers in the previous group are fully seated. On the other extreme, when λ is equal to 1, the time between calling groups to board is so short that the passengers in each group line up behind the previous group in the aisle.

Let AB represents the total aisle interferences time between groups, like previous sections,

$$AB = \lambda \sum_{i=1}^n \sum_{j=1}^6 \sum_{k=2}^G y_2 AB_{i,j,k}$$

Thus the total interferences time Q can be denoted as follows:

$$Q = TS + AW + AB$$

(3) Mathematical model

We can now summarize our multiple objective programming model as follows:

$$\text{minimize } Q \quad (3)$$

$$\text{minimize } T \quad (4)$$

Subject to:

$$S_{i,js,k} \geq x_{i,j,k} + \sum_{l=1}^k x_{i,s,l} - 1 \quad \forall i, \text{ and } k > 1 \quad (5)$$

$$AW_{i,j,k} \geq (6(i-1) + 4)x_{i,j,k} + \sum_{u=1}^i \sum_{v=1}^6 x_{u,v,k} - (6(i-1) + 5) \quad (6)$$

$$AB_{i,j,k} \geq 6(i)x_{i,j,k} + \sum_{u=1}^i \sum_{v=1}^6 x_{u,v,k-1} - 6(i) \quad (7)$$

$$TS = \alpha \sum_{i=1}^n \sum_{j=1}^m \sum_{k=2}^G (\sum_s y_1 S_{i,js,k}) \quad (8)$$

$$AW = \frac{1}{2} \beta \sum_{i=2}^n \sum_{j=1}^m \sum_{k=1}^G y_2 AW_{i,j,k} \quad (9)$$

$$AB = \lambda \sum_{i=1}^n \sum_{j=1}^m \sum_{k=2}^G y_2 AB_{i,j,k} \quad (10)$$

$$\sum_{k=1}^G x_{i,j,k} = 1 \quad \forall i, j \quad (11)$$

$$\sum_{i=1}^n \sum_{j=1}^6 x_{i,j,k} \geq M_1 \quad (12)$$

$$\sum_{i=1}^n \sum_{j=1}^6 x_{i,j,k} \leq M_2 \quad (13)$$

$$x_{i,j,k} \in \{0,1\} \quad \forall i, j, k \quad (14)$$

$$S_{i,js,k} \in \{0,1\} \quad \forall i, j, k \quad (15)$$

The objective function (3) attempts to minimize the total boarding time and total interferences time. The set of constraints (4) to (10) were explained extensively above. (11) ensures that each seat is assigned to only one group. The constraints (12) and (13) assign a minimum (M_1) and maximum (M_2) number of passengers to each group and finally (14) and (15) indicates that the decision variables are binary.

7 Computational results(for Outside-in system)

As T and Q are related to many factors, it's hard to obtain the optimal solution by general methods. So we use matlab to simulate them separately.

7.1 Simulation for T(boarding time)(See the program in the

appendix[5])

According to the idea introduced in the model, we simulate a series of values of time interval.

7.2 Simulation for $Q(\text{interference time})$ (See the program in the appendix[5])

Consider the Outside-in system which has no seat interference, where $TS = 0$ in the model. We only need to consider AW and AB .

Simulation steps are as follows:

We assume the curve, which we want to solve, is boarding-interval curve(BIC);

Step 1: We calculate the minimal boarding time by the simulation using the following steps. First we divide 150 people into 3 groups with equal number of people. Each group is arranged with a random order. Then we compute the boarding time according to this order for each group and then summarized them to obtain the whole boarding time. Assume now the time interval is 0, the boarding time obtained is 11.4386 min. Thus the average boarding time for each passenger is 4.75s. Obviously, the boarding time obtained is also 11.4386 min when the average boarding time for each passenger is less than 4.75s. This relation is described in fig3

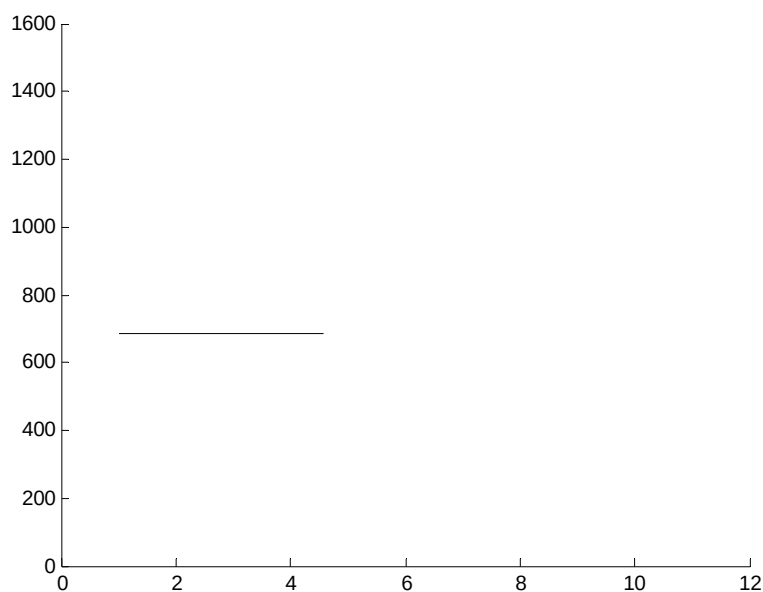


Figure 3

It is easy to see that the boarding time is $7 \times 150s = 17.5min$ when time interval is 7s,

since we assume the average time need for each passenger to put his or her luggage is 7s. Thus we have every reason to believe that the boarding time is $y \cdot 150$ s when time interval is y (s) when $y \geq 6$, which is also verified by our intensive simulation results. Now we obtain the fig4.

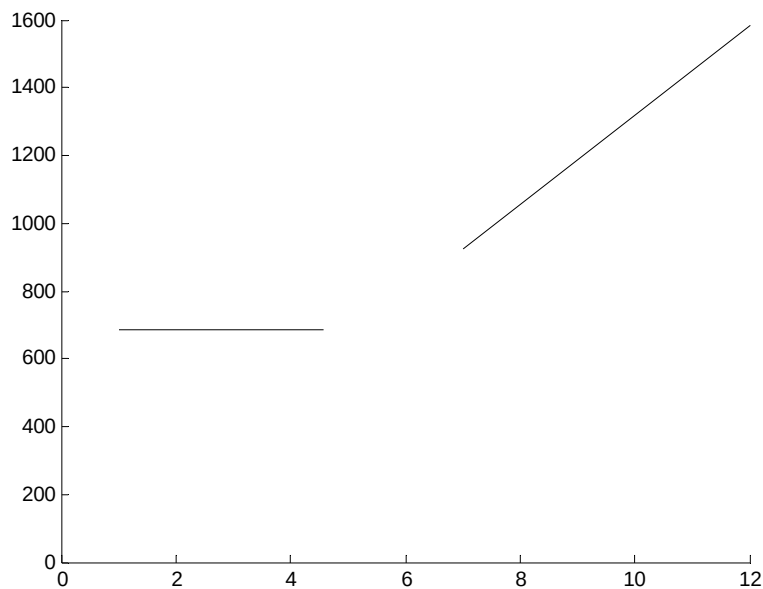


Figure 4

To finish the last step for whole BIV curve, the graph on the interval $[4.75, 6]$ can be obtained by curve fit technique. The BIV curve is given in fig 5

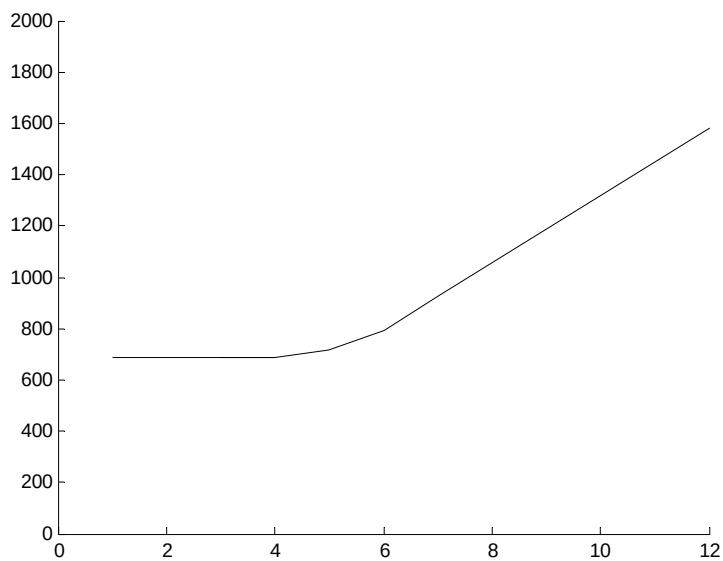


Figure 5

Now in a way similar to the above procedure we get the interference time curve as follows.(figure 6)

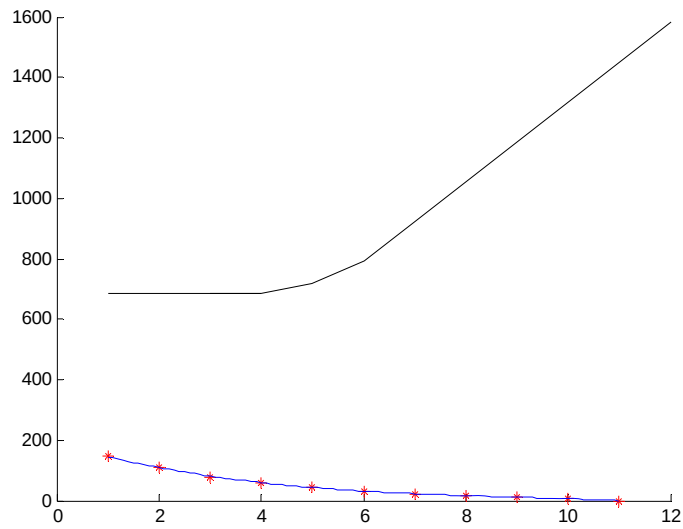


Figure 6

7.3 Simulation results

Small airplane (Airbus A320):

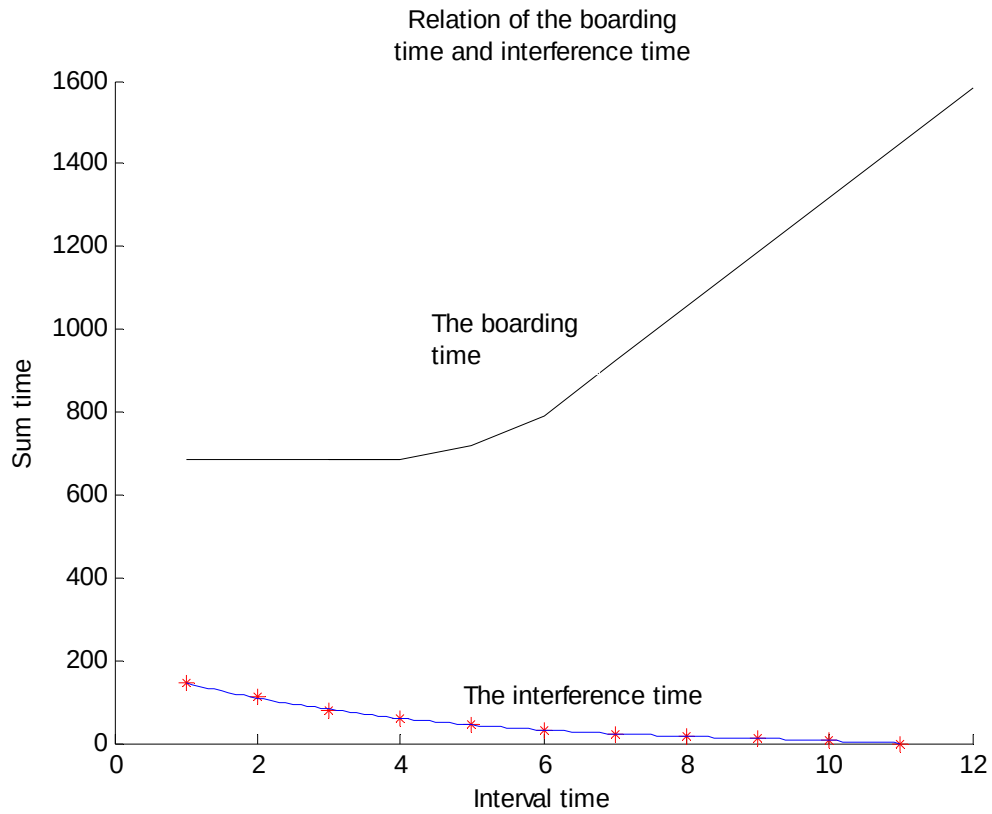


Figure 7 (Airbus A320) Relation of the boarding time and interference time

From the figure 7 we can see that the boarding time is the function increasing as the interval time, while the interference time is the function decreasing as time interval. The results are reasonable.

7.4 Determine the optimal time interval

Our objective function is to minimize Q and T . We give Q a weight ω_1 , give T a weight ω_2 . Then our objective function can be written as

$$\min \omega_1 Q + \omega_2 T$$

Here $\omega_1 + \omega_2 = 1$.

Step 1 From figure 7, we can fit two function whose variables are time interval. The boarding time function

$$T = \begin{cases} 686.316 & x < 4.75 \\ 2.1930 * x^3 - 10.895 * x^2 - 4.876 * x + 739.78 & 4.75 \leq x < 7 \\ 150 * x & 7 \leq x \leq 11 \end{cases}$$

The interference time function

$$Q = -0.1783 * x^3 + 4.8347 * x^2 - 48.84 * x + 189.1283$$

Step 2 Uniformization disposal

For different (ω_1, ω_2) , the optimal time interval is different, some conclusions are shown in the following table.(See the program in the appendix[4])

Table 7 (Airbus A320) Optimal time interval for different (ω_1, ω_2)

Interferences Time Weight ω_1	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
Boarding Time Weight ω_2	1	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1	0
Optimal Time Interval	4	4	4.2	4.5	4.9	5.2	5.6	7	7.3	11	11

Interferences represent the customers' benefit and boarding time can be seen as the company's profit . The best select is decided by the company's tendency . If the profit is much more important , the boarding time weight ω_2 is much larger than ω_1 . In reality , the profit plays very important role in a company ,and the company is also constrained by customers more or less .

8 Middle airplane(See the program in the appendix[3]):

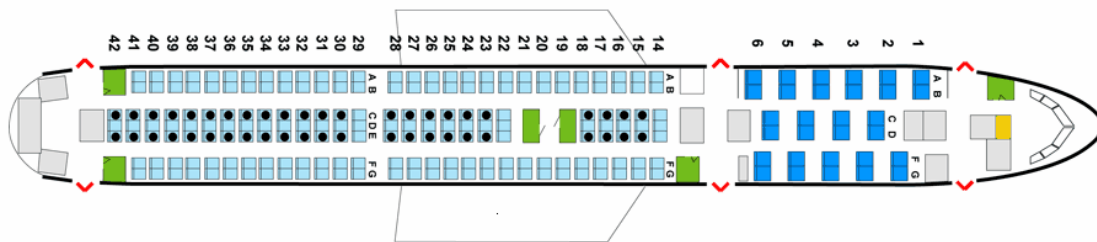


Figure 8 Boeing 747-400 airplane

According to 7.3 and 7.4, we obtain the following results for Figure 8.

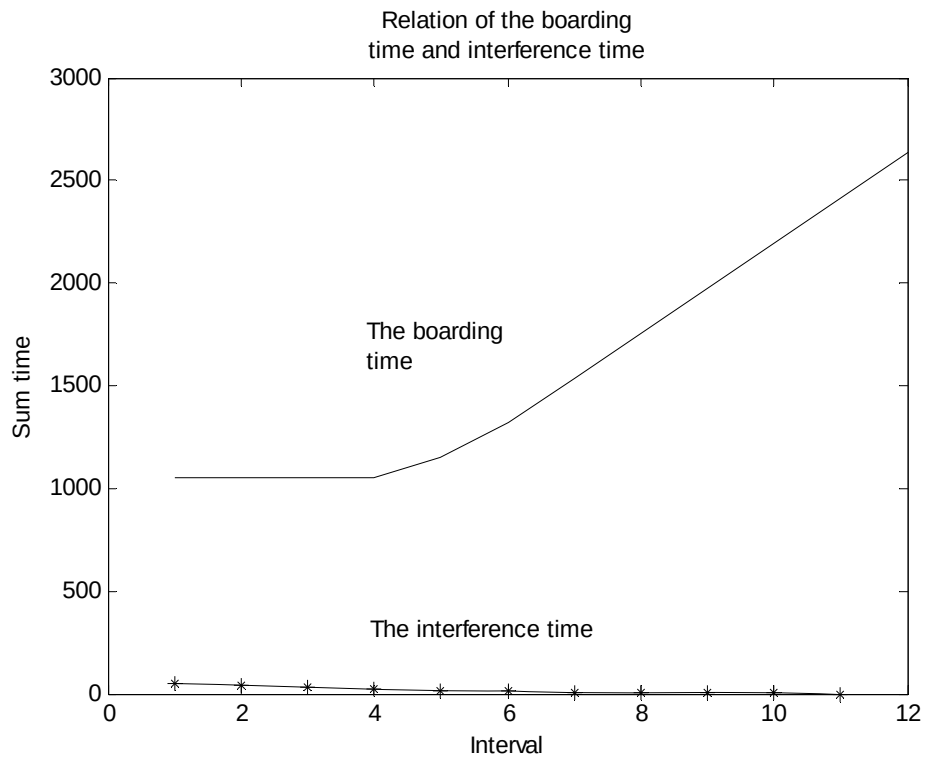


Figure 9 (Boeing 747-400) Relation of the boarding time and interference time

Table 8 (Boeing 747-400) Optimal time interval for different (ω_1, ω_2)

Interferences Time Weight ω_1	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
Boarding Time Weight ω_2	1	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1	0
Optimal Time Interval	3.5	3.7	3.9	4.2	4.6	5	5.6	6.4	7.9	11	11

9 Large airplane



Figure 10 A380 airplane

According to 7.3 and 7.4, we obtain the following results for Figure 10.

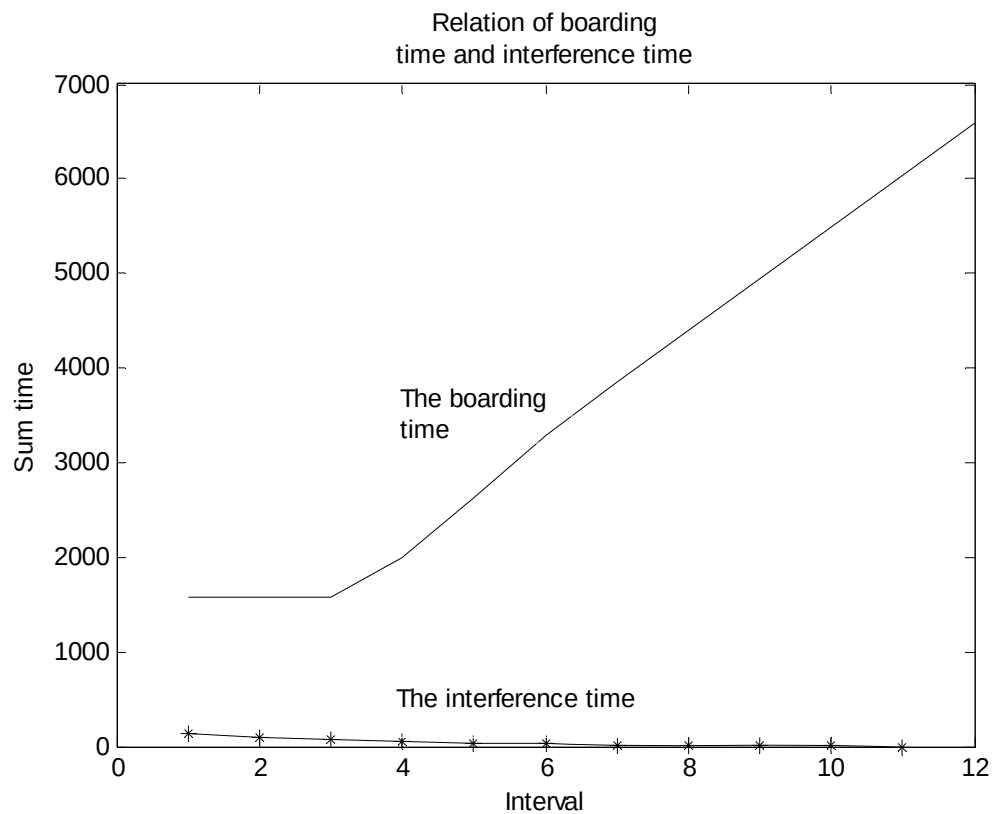


Figure 11 (A380) Relation of the boarding time and interference time

Large airplane

Interferences Time Weight ω_1	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.7	0.9	1
Boarding Time Weight	1	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1	0
Optimal Time Interval	2.5	2.7	2.9	3.2	3.6	4.1	4.9	7	8.4	11	11

Analysis:

- When Airlines consider passenger's satisfaction first, i.e. ω_1 approaches 1, the optimal interval of three types of airplanes approach 11s.
- When Airlines consider boarding time first, the optimal interval of three types of airplanes is much different:

$$\text{small airplane} > \text{middle airplane} > \text{large airplane}$$

10 Sensitivity Analysis

See Figure 7

- (1) Towards to the interrupt time; the longer is the time distance, the shorter is the interrupt time. Therefore, the time distance should be enough long.
- (2) For the boarding time; it's obvious that the boarding time is constant when the time distance is shorter than 4 seconds or so. However, when the distance time is becoming longer and longer, the boarding time prolong more and more quickly.

11 Strengths & Weaknesses

Strengths

- We simulate and analyze many boarding systems.
- We develop a multiple objective programming model which considers both boarding time and passengers' satisfaction
- We solve the model by using simulation and obtain reasonable results
- We obtain the optimal time interval

Weakness

- We don't consider the accidents.
- We don't consider the situation that the airplane is not 100% load.

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Appendix

[1] Reasonable analysis (simulate the four boarding systems presented) 1,Rand programme

```
%This programme is used to calculate the sum time of random seat.
clear
row_num = 6;           % the seat number of row
group_num = 4;         % the group number that we devided
row_speed = 1.5;       % the time of passing per row
one_time = 9;          % the time of between two passenger
bag_pro = 0.6;         % the probability of having luggage
bag_time = 7.1;        % the time of putting the luggage
seat_inter = 9.7;      % the time of seat interrupt
Sum_row = 23;          % the sum row of the plane
Sum_time = 0;          % the sum time of the boarding
count1 = 0;            % the interrupt count because of the bag and the seat
count2 = 0;            % the interrupt count because of the seat
group_sum = group_num * row_num;
A = randperm(group_sum); % the seat is random
for i = 2:group_sum
    for i1 = 1:i
        if (A(i1)+1 == A(i)) || (A(i1)+2 == A(i)) % occure the seat interrupt
            if rand <= 0.6
                Sum_time = Sum_time + bag_time + seat_inter;
                count1 = count1 + 1;
                break;
            else
                Sum_time = Sum_time + bag_time;
                count2 = count2 + 1;
                break;
            end
        end
    end
end
end
```

```

end
Sum_time = Sum_time-9*(count1 + count2 - 2);
count1;
count2;
count1/count2 % the ratio of having bag and no bag, it's better when it
                % approves to (60%)/(40%)
Sum_time = 22.5 + Sum_time/60;
Sum_time

```

[2] Minimum time of the four boarding procedure

1, Outside-in

% This programme is outside-in. It is used to calculate the time of
 % boarding one by one. That is, the time is 0 second not 9 seconds between
 % two passenger.

```

clear
row_sum = 23; % the sum row of the plane
row_speed = 1.5; % the time of passing per row
bag_time = 7.1; % the time of putting the luggage
seat_inter = 9.7; % the time of seat interrupt
Sum = 0; % the sum row of the plane
row_num = 6; % the seat number of row
for i = 1:1000
    time_sum = 0;
    i11 = 0;
    time_sum = time_sum + row_sum * row_speed;
    % the time of the passenger reaching the last row
    for i = 1:row_sum
        a(i) = rand; % The having luggage is random.
    end
    i = 1;
    A = randperm(row_sum);
    % The seat is random.
    while i < row_sum
        if a(i)<0.6
            for i1 = (i+1):row_sum
                if A(i1) < A(i)
                    time_sum = time_sum + bag_time;
                    % The bridge is jam.
                    i = i+1;
                    break;
                elseif a(i1) < 0.6
                    i = i+1;
                    break;
                end
            end
        end
    end
end

```

```

                                % The passenger have not the luggage
else
    i11 = i11+1;
end
end
else
    i = i+1;
end
i = i + i11;
i11 = 0;
end
if a(row_sum) < 0.6
    time_sum = time_sum + bag_time;
    % the last passenger is considered.
end
Sum = Sum + time_sum;
end
Sum = Sum*6/60/1000;
Sum

```

2, Reverse-pyramid

% The programme is reverse-pyramid. It is used to caculate the time of
 % boarding one by one. That is, the time is 0 second not 9 seconds between
 % two passenge.

```

clear
row_sum = 14;           % the divided group number
row_speed = 1.5;        % the time of passing per row
bag_time = [3.1,5.1,7.1,9.1,11.1]; % the time of puting the luggage
seat_inter = 9.7;       % the time of seat interrupt
Sum = 0;                % the sum time of the boarding
row_num = 6;            % the seat number of row
time_sum = 0;
for i = 1:1000
    time_sum = 0;
    for j = 1:5
        i11 = 0;
        time_sum = time_sum + row_sum * row_speed;
        % the time of the passenger reach the last row
    for i = 1:row_sum
        a(i) = rand;
    end
    i = 1;
    A = randperm(row_sum);
    while i < row_sum

```



```

        if a(i)<0.6
            for i1 = (i+1):row_sum
                if A(i1) < A(i)
                    time_sum = time_sum + bag_time(j);
                    i = i+1;
                    break;
                elseif a(i1) < 0.6
                    i = i+1;
                    break;
                else
                    i11 = i11+1;
                end
            end
        else
            i = i+1;
        end
        i = i + i11;
        i11 = 0;
    end
    if a(row_sum) < 0.6
        time_sum = time_sum + bag_time(j);
    end
    Sum = Sum + time_sum;
end
end
Sum = Sum/60/1000;
Sum

```

3, U1

% The programme estimate the time of U1. Because the simulation of U1 is very complex

%, so we can estimate the time of U1. The average of 10000 times is the % boarding time.

```

clear
bag_time=[3.1,7.1,11.1];           % the time of puting the luggage
row_sum = 23;                      % the sum row of the plane
seat_inter = 9.7;                  % the time of seat interrupt
Sum = 0;                           % the sum time of the boarding
bag_pro = 0.6;                     % the probability of having luggage
row_num = 6;                       % the seat number of row
group = [74,46,18];                % the group number that we divided
count = 0;
a = rand(1,group(1));
A = randperm(group(1));

```

```
for i = 1:10000
    % estimate the time of the first group
    for i = 1:group(1)
        if A(i) > 42
            Sum = Sum + bag_time(1);
            count = i;
            break;
        end
    end
    if a(count+1) < 0.6
        Sum = Sum + bag_time(1);
    end
    for i = (count+1):group(1)
        if A(i) < 42
            if a(i) < 0.6
                Sum = Sum + seat_inter + bag_time(1);
            end
        end
    end
    % estimate the time of the second group
    b = rand(1,group(2));
    B = randperm(group(2));
    for i = 1:group(2)
        if A(i) > 28
            Sum = Sum + bag_time(1);
            count = i;
            break;
        end
    end
    if a(count + 1) < 0.6
        Sum = Sum + bag_time(2);
    end
    for i = (count+1):group(2)
        if A(i) < 28
            if a(i) < 0.6
                Sum = Sum + seat_inter + bag_time(2);
            end
        end
    end
    % estimate the time of the third group
    c = rand(1,group(3));
    C = randperm(group(3));
    for i = 1:group(3)
        if c(i) < 0.6
```

```

        Sum = Sum + bag_time(3);
    end
end
end
Sum = Sum/60/10000;
Sum
4, U2
% The programme estimate the time of U2. Because the simulation of U2 is very
complex
% , so we can estimate the time of U2. The average of 10000 times is the
% boarding time.
clear
bag_time=[3.1,7.1,11.1];    % the time of puting the luggage
row_sum = 23;                % the sum row of the plane
seat_inter = 9.7;           % the time of seat interrupt
Sum = 0;                     % the sum time of the boarding
bag_pro = 0.6;               % the probability of having luggage
row_num = 6;                 % the seat number of row
group = [94,32,12];          % the group number that we divided
count = 0;
a = rand(1,group(1));
A = randperm(group(1));
for i = 1:10000
    % estimate the time of the first group
    for i = 1:group(1)
        if A(i) > 72
            Sum = Sum + bag_time(1);
            count = i;
            break;
        end
    end
    if a(count+1) < 0.6
        Sum = Sum + bag_time(1);
    end
    for i = (count+1):group(1)
        if A(i) < 72
            if a(i) < 0.6
                Sum = Sum + seat_inter + bag_time(1);
            end
        end
    end
    end
    b = rand(1,group(2));
    B = randperm(group(2));
    % estimate the time of the second group

```

```

for i = 1:group(2)
    if A(i) > 20
        Sum = Sum + bag_time(1);
        count = i;
        break;
    end
end
if a(count + 1) < 0.6
    Sum = Sum + bag_time(2);
end
for i = (count+1):group(2)
    if A(i) < 12
        if a(i) < 0.6
            Sum = Sum + seat_inter + bag_time(2);
        end
    end
end
% estimate the time of the third group
c = rand(1,group(3));
C = randperm(group(3));
for i = 1:group(3)
    if c(i) < 0.6
        Sum = Sum + bag_time(3);
    end
end
end
Sum = Sum/60/10000;
Sum

```

[3] The minimum the middle airplane

% This programme is Out-in for the midsize, whose number is 220. It is used to calculate the time of boarding one by one. That is, the time is 0 second not 9 seconds between two passengers.

```

clear
row_sum = 14; % the boarding number of per group
row_speed = 1.5; % the time of passing per row
bag_time = [3.1, 7.1, 11.1]; % the time of putting the luggage
seat_inter = 9.7; % the time of seat interrupt
Sum = 0; % the sum time of the boarding
row_num = 7; % the people number of every row
people_num = 220; % the sum number of the plane
r = 0.001;

```

```

for i = 1:1000
    time_sum = 0;
    i11 = 0;
    for j = 1:3
        time_sum = time_sum + 1.5 * row_sum * row_speed;
        for i = 1:row_sum
            a(i) = rand;
        end
        i = 1;
        A = randperm(row_sum);
        while i < row_sum
            if a(i) < 0.6
                for i1 = (i+1):row_sum
                    if A(i1) < A(i)
                        time_sum = time_sum + bag_time(j);
                        i = i+1;
                        break;
                    elseif a(i1) < 0.6
                        i = i+1;
                        break;
                    else
                        i11 = i11+1;
                    end
                end
            else
                i = i+1;
            end
            i = i + i11;
            i11 = 0;
        end
        if a(row_sum) < 0.6
            time_sum = time_sum + bag_time(j);
        end
    end
    Sum = Sum + time_sum;
    time_sum = time_sum + 1/2*time_sum;
end
Sum = Sum * 2/60/1000;
Sum = Sum * 2 + r * people_num

```

[4] Optimal time interval for different (ω_1, ω_2)

% The programme is used to fit the curve

```

axis([0 12 0 160])
hold on;
x = 1:11;
y = [145.1620 110.6630 77.6040 60.1660 46.7160 30.7960 22.2400 16.7790
11.0040 5.4290 0];
p = polyfit(x,y,3)
x1 = 1:0.1:11
y1 = polyval(p,x1);
plot(x,y,'*r',x1,y1,'-b')

```

% The programme is used to design good project for the small airplane.

```

clear
sum_time = 11.4386; % the minimal time
sum_people = 132; % the people number
for w = 0:0.1:1; % weight
    y = 10000;
    for x = 1:0.1:11;
        y1 = -0.1783*x.^3 + 4.8347*x.^2 - 48.8443*x + 189.1283;
        y1 = y1/144.9404;
        if x < floor(sum_time*60/sum_people)-1
            y2 = sum_time*60;
        elseif x < 7
            y2 = 2.1930 * x.^3 - 10.895*x.^2 - 4.876*x + 739.78;
        elseif x <= 11
            y2 = x * 123;
        end
        y2 = (y2 - sum_time * 60)/(11*123 - sum_time * 60);
        if w*y1 + (1-w)*y2 < y
            x1 = x;
            y = w*y1 + (1-w)*y2;
        end
    end
end
x1
end

```

[5] The boarding time and the interference time

% The programme is calculate the interrupt time and draw the figure.

```

clear
bag_time = [3,7,11];
group_num = 46;
bag_pro = 0.6;

```

```

Sum = 0;
for time = 1:12
    Sum1(time) = 0;
    for t = 1:1000
        for i = 1:3
            a = rand(1,group_num);
            A = randperm(group_num);
            for i1 = 1:group_num
                if a(i1) < bag_pro
                    j = 1;
                    A1(j) = A(i1);
                    for t = (i1 + 1):group_num
                        if a(t) > bag_pro
                            j = j+1;
                            A1(j) = A(t);
                        else
                            break;
                        end % if
                    end % for
                    for j1 = 1:j-1 % calculate the time of the group
                        if A(j1+1) < A(1)
                            if bag_time(i)-j1 * time > 0
                                Sum = Sum + bag_time(i)-j1 * time;
                            end
                        end
                    end % for
                end % if
            end % for
        end % for(i = 1:3)
    end % for t= 1 :1000
    Sum1(time) = Sum/1000;
    Sum = 0;
end
x = 1 : 12;
y = Sum1(x);
plot(x,y,'-*k')
hold on;

```

%The programme is the boarding time and draw the figure.

```

clear
sum_time = 11.4386 * 60;
sum_num = 132;
x = [1,floor(sum_time/132)-2];
y = [sum_time,sum_time];

```

```

plot(x,y,'-k');
hold on;
x = [7,12];
y = [7*132,12*132];
plot(x,y,'-k');
x = [3,4,6,7];
y = [floor(sum_time/132),floor(sum_time/132),6*132,7*132];
eq1 = sym('a*4^3 + b*4^2 + c*4 + d = floor(sum_time)');
eq2 = sym('a*3^3 + b*3^2 + c*3 + d = floor(sum_time)');
eq3 = sym('a*6^3 + b*6^2 + c*6 + d = 6*132');
eq4 = sym('a*7^3 + b*7^2 + c*7 + d = 7*132');
[a,b,c,d] = solve('a*4^3 + b*4^2 + c*4 + d = 11.4386*60','a*3^3 + b*3^2 + c*3 + d = 11.4386*60','a*6^3 + b*6^2 + c*6 + d = 6*132','a*7^3 + b*7^2 + c*7 + d = 7*132')
x = 3:1:7
y = 2.1930 * x.^3 -10.895*x.^2 -4.876*x+ 739.78
plot(x,y,'-k')

```